Cancellable Biometric Template Generation Using Vector Quantization and Hill Cipher

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Abstract—This research proposed a new method to improve the security of encrypted biometric systems by combining vector quantization with Hill ciphers. The method involves quantizing biometric data into vectors, simplifying the representation while preserving critical information, followed by encryption using a Hill cipher. This technique generates multiple secure templates from the same biometric input, providing resilience against attacks. In addition, the reversibility of the Hill cipher allows regeneration of the original biometric data under permitted con- ditions. The proposed system offers improved privacy, security and flexibility for modern biometric systems and demonstrates robust performance.

Index Terms—Cancellable biometrics, vector quantization, Hill cipher, biometric security, template protection, cryptography

I. Introduction

Biometric technology refers to the automatic identification and identification of individuals based on their physical and behavioral characteristics [1]. This innovative field has gained immense popularity owing to its ability to provide secure and efficient alternative solutions for identification purposes over traditional methods like alphanumeric passwords or PINs.

Biometric identifiers are distinctive and measurable traits used to recognize and describe individuals. These identifiers are typically divided into two categories:

- Physical Biometrics: These encompass anatomical or physiological elements such as fingerprints, facial patterns, DNA, palm veins, hand geometry, iris structure, retinal patterns, and even the distinct scent of a person. Since these attributes are relatively constant over time, they serve as reliable indicators for identification.
- Behavioral Biometrics: This category includes characteristics associated with the patterns of behavior of an individual, including but not limited to voice recognition, gait analysis, keystroke dynamics, and signature style. These traits may vary somewhat due to a multitude of factors, including changes in emotional state, physical conditions, or the environment.

Biometric systems work primarily by capturing patterns of biometric characteristics, such as fingerprints or voice recordings [2]. This sample is processed and transformed into a digital representation and then matched against a pre-existing database of biometric data [3]. If the comparison yields a high enough similarity score, the identity of the individual is confirmed.

The deployment of biometric technology spans various applications, from security and law enforcement to consumer electronics, where they enhance accessibility and convenience. The unique advantages of biometrics include heightened se- curity due to the difficulty of duplicating biometric traits, the elimination of the need to remember passwords or carry ID cards, and improved accuracy in identity verification, which mitigates the risk of unauthorized access or fraudulent activities [4].

Despite these benefits, the implementation of biometric tech- nology is not without challenges, particularly regarding con- cerns about privacy, security of biometric data, and the poten- tial for abuse [5]. Ethical considerations are paramount, as the integration of this technology into societal norms necessitates a delicate balance between individual privacy rights and col- lective security requirements. As biometric systems become more pervasive, these issues are increasingly becoming the subject of public discourse and regulatory attention.

A. The Principle of Cancellable Biometrics

The principle of reversing biometrics is to transform biometric information into a new form that can be removed and replaced when modified. The raw biometric data never leaves the secure area of the device and only the modified version is stored or transmitted. This ensures that even if the stored biometric data is compromised or stolen, the actual biometric data cannot be reproduced, thus protecting the user's privacy and security, as shown in the Fig. 1 [6].

II. Literature Review

Cancellable biometrics is an emerging field aimed at enhancing the security and privacy of biometric systems. Traditional biometric systems are vulnerable to attacks, as biometric traits are irrevocable once compromised. To address this, cancellable biometrics transforms biometric data into a secure, non-invertible format, allowing for revocation and reissuance in case of data breach. This technique not only preserves the integrity of biometric authentication but also ensures user privacy through various methods such as vector quantization, encryption, and hashing. The advancement in this field is crucial for developing robust and secure biometric systems suitable for diverse applications.

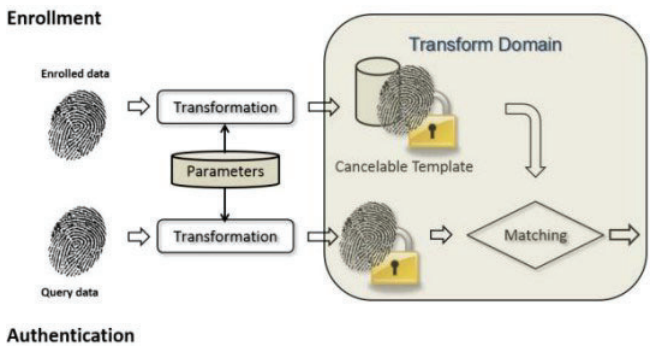

Fig. 1. Cancellable biometric. [6]

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A. Encryption-based Cancellable Biometrics

- Bansal et al. [7] introduced a CBTP strategy combining FPE with Bloom filters, achieving noteworthy authentication accuracy with very low FRR and FAR on the IITD- CASIA dataset.
- Kausar et al. [8] developed the ICBC method for protecting patient healthcare data on smart cards, utilizing an SKC approach paired with the FC method for robust encryption.

B. Visual Secret Sharing and Genetic Algorithms

- Manisha et al. [9] proposed the CBT-RBXOR method, leveraging VSS to create secure cancellable templates with excellent quantitative results.
- El-Shafai et al. [15] explored a GA-based cancellable bio metric model, demonstrating optimal performance even in noisy conditions.

C. Fourier Transform and Multimodal Techniques

- Alarifi et al. [10] put forth a PTFT-based cancellable biometric recognition system, showcasing robustness and security across multiple datasets.
- Lee et al. [11] envisioned a transformative technique using multimodal biometrics, yielding high recognition rates and practically non-existent FAR.

D. Deep Learning and Robust Feature Extraction

- Sedik's team [12] suggested a deep learning-based cancellable biometric system that processes encrypted modalities with high efficiency.
- Walia et al. [13] introduced a secure multimodal technique, demonstrating improved resilience to presentation attacks and enhanced privacy measures.

E. Local Ranking-Based Iris-Code Protection

- Ouda et al. [14] devised a method for iris-code protection, affirming its non-reversibility and resistance to sophisti- cated attacks.

III. Proposed Work And Methodology

Retinal recognition is ideal for secure biometric systems because it is accurate and tamper-proof. In this paper, we present a retinal image-based biometric authentication system that ensures privacy and security of biometric information through encryption while maintaining data integrity. The IDRID dataset is used as the basis for reti- nal image detection in the proposed method. The system operates in two stages: enrollment and authentication. During enrollment, the retinal image is converted to a sin- gle channel through grayscale transformation to improve performance. Segmentation is performed by seed-based random vectors that divide the image into several re- gions to prepare for feature extraction. Log-Gabor filters are used to extract high-frequency features from retinal segments that are necessary to identify unique biometric patterns. The output of the filter is mathematically defined as follows:

$$G(f,\theta) = \exp\left(-\frac{(\log(f/f_0))^2}{2\sigma_f^2}\right)\exp\left(-\frac{(\theta-\theta_0)^2}{2\sigma_\theta^2}\right) \quad (1)$$

where f represents the frequency, θ denotes orientation, f0 and θ0 are the central frequency and orientation, and σf , σθ control the bandwidth.

After feature extraction, vector quantization is applied using K-means clustering. This process transforms the feature set into cancellable codewords. Given a set of n feature vectors x1, x2, . . . , xn, the quantized codebook C = {c1, c2, . . . , ck} minimizes the distortion function:

$$D = \sum_{i=1}^{n} \min_j \|\mathbf{x}_i - \mathbf{c}_j\|^2 \quad (2)$$

Then use Hill cipher to create an encrypted biometric template defined by the matrix equation:

$$C = K \cdot P \pmod m \quad (3)$$

where C is the ciphertext, P is the plaintext (the quantized image), K is the encryption key matrix, and m is the modular base. The use of matrix-based encryption ensures robustness against cryptographic attacks.

In the verification phase, the same preprocessing steps (grayscale conversion, segmentation, feature extraction, and vector quantization) are applied to the input images. The generated model is encrypted using the Hill cipher and compared with the encrypted model stored in the file. A successful campaign verifies the user's identity.

A. IDRID Dataset

Initially used for diabetic retinopathy research, the IDRID dataset is well suited for retinal biometry due to its high-resolution images and detailed descriptions. In this case, the structure of the captured retinal vessels is unique to each person and plays an important role in ensuring the smoothness and rigidity of the body.

B. Feature Extraction Using Log-Gabor Filter

Grayscale conversion simplifies image processing by representing the image in a single intensity channel. The log-Gabor filter extracts detailed texture information, which is crucial for retinal biometrics. It is mathematically defined as mentioned earlier in equation 1, where it filters specific frequency and orientation bands, highlighting distinctive retinal patterns.

C. Vector Quantization for Cancellable Biometrics

K-means clustering, used for vector quantization, transforms the extracted features into cancellable templates, ensuring that the biometric data can be replaced with- out compromising security. The clusters minimize intra- cluster variance, creating a more compact and cancellable biometric template.

D. Encryption with Hill Cipher

The Hill cipher provides encryption using matrix multiplication. The biometric template (a vector form of the quantized features) is encrypted with a key matrix K, ensuring that even minor changes in the template result in a completely different encrypted output. The decryption process involves using the inverse key matrix K−1:

$$P = K^{-1} \cdot C \pmod m \quad (4)$$

where P is the decrypted biometric template.

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E. System Flow

As shown in Fig. 2, the comprehensive processes involved in both the enrollment and authentication phases of the biometric system are represented.

IV. Result And Discussion

The advent of biometric systems has revolutionized personal authentication by leveraging unique human features for identification. However, the irrevocable nature of bio- metric traits raises significant privacy concerns, as once compromised, they cannot be reissued or revoked like traditional authentication tokens. To mitigate such risks, we leverage biometric technology to provide solutions that protect the integrity of biometric data and user privacy.

Cancellable biometrics employ transformation techniques to convert original biometric data into a new, secure, and revocable representation. Among these techniques, vector quantization plays a pivotal role, providing an efficient and effective method for generating protected templates. Vector quantization discretizes the feature space into a finite number of regions, known as clusters, and repre- sents each region by its centroid. The process is inherently irreversible, as it introduces quantization error, preventing the original data from being precisely reconstructed from the centroids alone.

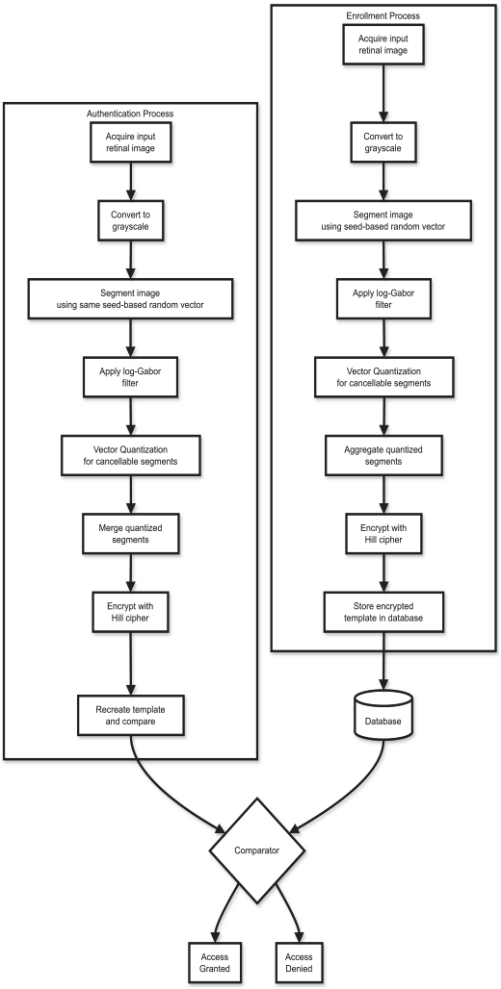

Fig. 2. flowchart for the cancellable biometric system based on retinal image. The above flow diagram is divided into two segments, shows the enrollment process on the left hand side and the authentication process on the right hand side.

This experimental analysis explores the application of vector quantization in the creation of cancellable bio- metric templates. By transforming biometric data into a quantized space, we hypothesize that the resulting templates will not only be secure and private but also uphold the performance standards necessary for practical biometric systems. The experimental setup involves pro- cessing biometric images from a public dataset, applying vector quantization, and evaluating the irreversibility and uniqueness of the generated templates.

Through this analysis, we anticipate demonstrating that vector quantization serves as an effective tool for produc- ing cancellable biometric templates, contributing to the advancement of secure and privacy-preserving biometric authentication systems.

A. Gray Scale Conversion

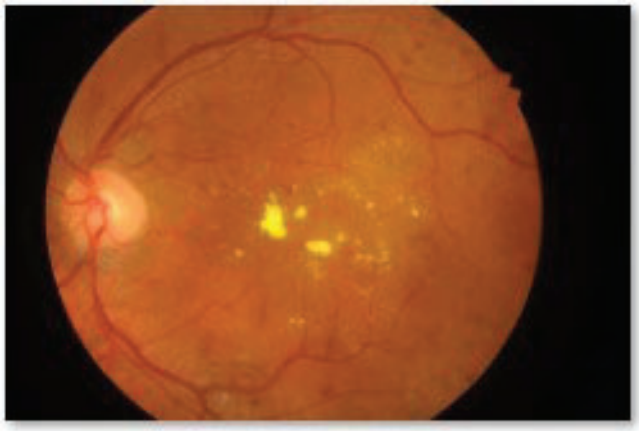

Fig. 3. Original Image [16]

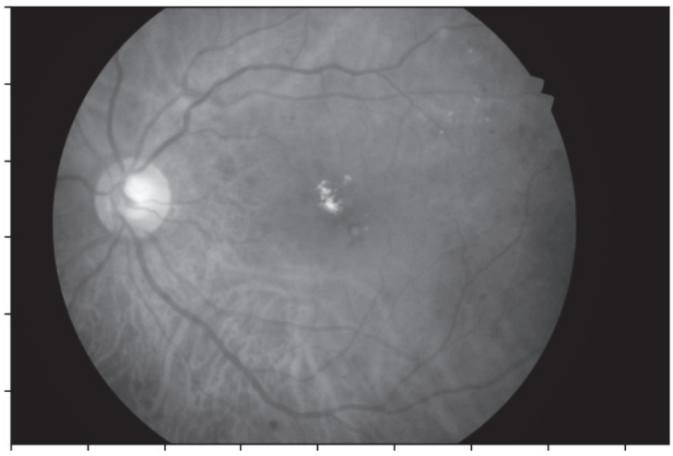

Fig. 4. Gray Scale Image

Fig. 3 shows the original image, which contains all the color information as captured by the imaging device. In contrast, Fig. 4 displays the grayscale version of the same image. The grayscale image is generated by removing the color information and representing the image in shades of gray, which typically range from black to white. This conversion can be useful in various image processing tasks where color is not a critical factor.

B. Splitting of Gray Scale Image

Fig. 5 illustrates the process of splitting an image into its grayscale components. This technique involves convert- ing

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the original color image into multiple grayscale im ages, each representing different aspects or regions of the original image. By examining these individual grayscale components, it is possible to analyze the image's struc- tural and textural features more effectively. This method is often used in image processing and computer vision applications to enhance feature extraction and improve the performance of various algorithms.

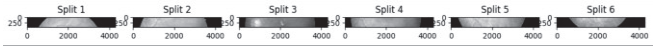

Fig. 5. Gray Scale Split

C. Feature Extraction Using Log-Gabor

Fig. 6. Feature Extraction

Fig. 6 demonstrates the process of feature extraction using Log Gabor filters. Log Gabor filters are particularly effective in capturing texture and edge information in images due to their logarithmic frequency response. Unlike traditional Gabor filters, Log Gabor filters have no DC component, and their frequency response is more adaptable to natural image statistics, making them more suitable for analyzing a wide range of frequencies.

In the figure, we can see the original image on the left and the extracted features on the right. The extracted features highlight important structural details and textures within the image, which are crucial for various image processing tasks such as object recognition, image classification, and biometric analysis. The use of Log Gabor filters allows for enhanced sensitivity to both high and low-frequency components, providing a comprehensive representation of the image's features.

This technique is widely used in computer vision and im- age analysis applications where robust feature extraction is necessary. By capturing both fine and coarse details, Log Gabor-based feature extraction facilitates improved performance in subsequent image processing stages.

D. Vector Quantization

Fig. 7. Vector Quantization of Each Image

Fig. 7 illustrates the process of vector quantization for a single image. Vector quantization is used in signal processing and image compression to summarize the data by erasing some of the individual pixel values of an image but retaining the general characteristics. The original image is shown to the left of the figure and the quantised version is shown on the right side. In vector quantization, the image is broken up into blocks and represented by a vector number in a pre-written book. This code list is created through a learning process in which the code vectors are optimized to best represent the image block. Quantized images carry forward the essence of reducing detail and complexity since continuous ranges of pixel values are mapped to a limited set of representative values. However, it can be noted that most structural and textural features of the original image remain intact, and this proves that vector quantization is very efficient in compressing data with the retention of image integrity. It has vast applications in image compression algorithms as well as in pattern recognition problems.

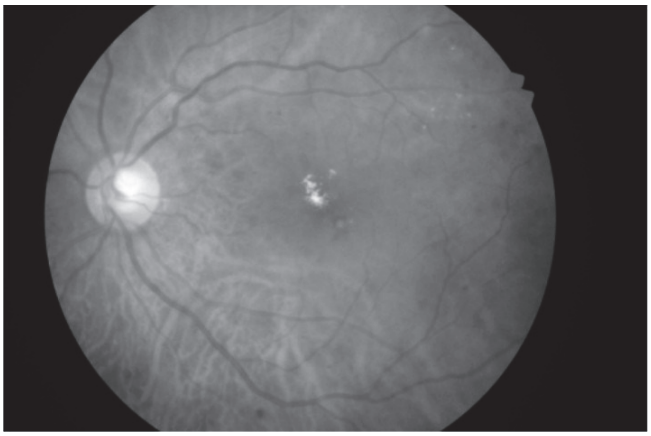

Fig. 8. Merge of Quantized Image

Fig. 8 shows the result of merging multiple quantized images into a single composite image. In image pro- cessing, quantization involves reducing the complexity of an image by mapping a large set of input values to a smaller set while preserving essential properties. The figure shows the integration of multiple quantized images into a single representation. This fused image extracts fusion information from the individual quantized segments and comprehensively represents the structural and textural features of the original image. Combining quantized images can enhance image data analysis and processing. This technique could be useful in applications such as image compression and biometric analysis, where reducing data size while retaining important information is important.

E. Encryption Using Hill Cipher

Fig. 9 shows the result of applying the Hill cipher algorithm to an image. The Hill cipher is a multi-word substitution cipher based on linear algebra. It is suitable for encrypting image data because it uses matrix multipli- cation to convert blocks of plaintext into ciphertext. The image shows an encrypted version of the original image. The encryption process involves splitting the image into small blocks of pixel values, which are then processed into vectors. By multiplying these vectors by a reversible key matrix, an encrypted image block is created.

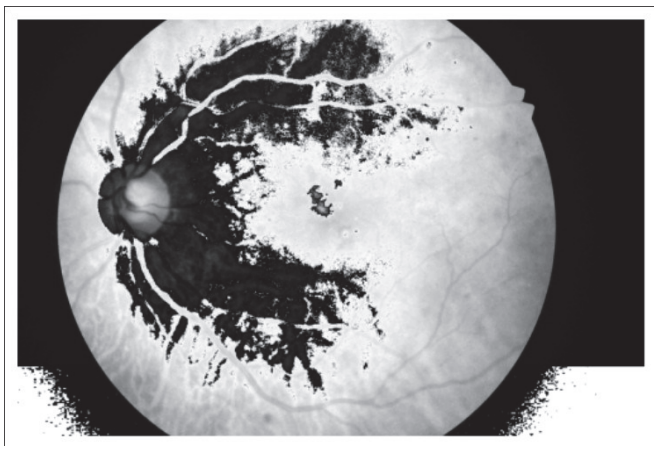

Fig. 9. Hill Cipher Encrypted Image

The resulting encrypted image appears to be significantly different from the original, hiding the original visual in- formation by 'transforming' the pixel values. The strength of the Hill cipher lies in its ability to propagate the characteristics of the plaintext throughout the ciphertext, providing a high

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level of security. This makes it difficult to obtain the original image without the corresponding decryption key matrix. Encrypting an image using the Hill cipher protects the image content from unauthorized access, as it cannot be easily deciphered by unauthorized persons.

F. Template of Same Image Comparison

Fig. 10. Difference Between Two Templates of the Same Image

Fig. 10 shows the difference between two templates derived from the same image using different random vectors. A template refers to a processed version of an image generated through a specific transformation or feature extraction method, where different seed vectors lead to variations in the resulting templates.

The figure shows how a changing random seed vector affects the appearance of the templates. Using different seed vectors introduces subtle or significant changes to the transformation process, resulting in distinct templates that exhibit noticeable differences due to the randomness involved.

This technique is useful in applications where changes are required for security or other reasons. For example, in a biometric system, using different genes to create multiple models from the same biometric data can improve secu- rity by making it more difficult for an attacker to copy or guess the actual model. Similarly, when augmenting data for machine learning, such changes can improve the robustness and generalization of the model.

V. Conclusion

In this paper, we present a novel method for decrypting biometric systems using vector quantization combined with Hill ciphers. Our method converts biometric infor- mation into quantitative form and guarantees irreversibil- ity, which is important for self-defense development. Other implementations of Hill ciphers further enhance the security of biometric data and prevent unauthorized ac- cess. Our experimental results show that this combination can effectively protect biometric data while maintaining important features required for practical authentication and identity verification.

VI. Future Scope

Future research should focus on optimizing vector quan- tization techniques to improve data compression, irre- versibility, and performance preservation. Looking for advanced algorithms or machine learning techniques can improve performance. Also, examining the robustness of the theory against various attacks, including reverse engi- neering and high-level escalation, is important for further security development. As biometric databases expand, research into ways to speed up the process and encryption (e.g., through benchmarking or hardware acceleration) will be important for practical use. Integration of non- extractable biometric systems with multimodal biometric technology can increase accuracy and reliability. Finally, real-world application and user validation studies are needed to evaluate performance and compliance with privacy laws and standards.

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